

NON-INTERACTING TANKS

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Objective:

The aim of this experiment is to study the dynamic response of the second order critically damped system consisting of a set of non-interacting tanks to a step change in the input. The experiment aims to determine the system's time constants, analyze its transient behavior, and compare the experimental results with theoretical predictions.

Theory and Background:

Fluid flow and level control are fundamental to ensuring the reliability and efficiency of many industrial and chemical operations. A non-interacting tank system provides a simplified yet powerful model to study such dynamics. It typically comprises two tanks arranged in series, where the effluent from the first tank is discharged into the atmosphere before entering the second. This setup ensures that the behavior of each tank is independent i.e., changes in one tank do not influence the other allowing the system to be treated as two first-order processes connected in sequence. The term "non-interacting" indicates that the dynamics of each tank are independent meaning that the liquid level in one tank does not influence the flow behavior in the other.

When a step change is introduced in the input flow rate, the liquid levels in both tanks gradually adjust until a new steady state is achieved. This dynamic behavior is governed by the time constants of each tank, which are determined by their cross-sectional areas and flow resistances. In a non-interacting tank system, where each tank responds independently, the analysis becomes straightforward. The system's overall response depends on the relationship between the individual time constants, which influences whether the response is overdamped or critically damped. In the critically damped case when the time constants of both tanks are approximately equal the system returns to equilibrium without oscillations and in the shortest possible time, ensuring a smooth and efficient transition.

Non-interacting tank systems are widely used as dynamic models in industries such as chemical manufacturing, water treatment, and thermal power generation. They play a crucial role in maintaining stable operating conditions such as regulating reactant feeds in chemical reactors, controlling water flow through filters in treatment plants, or managing boiler water levels to avoid thermal stress and system failures. Understanding their behavior is vital for designing responsive and control strategies in such applications.

Basic terms to know:

1. *Time Constant (τ): It shows how fast a system reacts to input changes.*
2. *Critically Damped System: A second order system that goes back to its steady states without any bouncing or oscillations*
3. *Resistance (R): The blockage or opposition to fluid flow, usually caused by the outlet tube of the tank.*
4. *Steady State: When the system values, like liquid level, stay constant over time.*

5. *First order system* : A system where the rate of change of the output variable linearly depends on its current state.
6. *Second Order System*: A system where the output depends both on its current value and how fast it is changing.

Derivations:

Mass balance:

- For tank 1:

$$\circ A_1 \frac{dh_1}{dt} = q - q_1 = q - \frac{h_1}{R_1}$$

■ Where:

- A_1 is the cross – sectional area of tank 1
- h_1 is the height of liquid in tank 1
- q is the total inflow into tank 1
- $q_1 = \frac{h_1}{R_1}$ (using the concept of hydraulic resistance)
- R_1 is the resistance in Tank 1

- For tank 2:

$$\circ A_2 \frac{dh_2}{dt} = q_1 - q_2 = \frac{h_1}{R_1} - \frac{h_2}{R_2}$$

■ Where:

- A_2 is the cross – sectional area of tank 2
- h_2 is the height of liquid in tank 2
- q_1 is the total inflow into tank 2 $= \frac{h_1}{R_1}$
- $q_2 = \frac{h_2}{R_2}$ (using the concept of hydraulic resistance)
- R_2 is the resistance in Tank 2

Flow Resistance relationship:

Assuming linear resistance to the flow, we get:

$$q_1 = \frac{h_1}{R_1} \text{ and } q_2 = \frac{h_2}{R_2}, \text{ where:}$$

- h_1 is the height of liquid in tank 1
- h_2 is the height of liquid in tank 2
- R_1 is the resistance in Tank 1
- R_2 is the resistance in Tank 2

- q_1 is the Output flow rate from Tank 1 (and input to Tank 2)
- q_2 is the Output flow rate from Tank 2

Transfer Function:

Assuming the initial conditions to be 0, and taking laplace transform of mass balance equations we get:

$$\frac{H_1(s)}{Q(s)} = \frac{R_1}{A_1 s R_1 + 1} = \frac{1}{A_1 s + 1/R_1}$$

$$\frac{H_2(s)}{H_1(s)} = \frac{R_2}{A_2 s R_2 + 1} = \frac{1}{A_2 s + 1/R_2}$$

The overall system would be: $\frac{H_2(s)}{Q(s)} = \frac{1}{A_1 s + 1/R_1} \times \frac{1}{A_2 s + 1/R_2}$

This confirms that the system behaves as a second-order critically damped system when the damping coefficient $\xi = 1$

Time constants and Damping coefficient:

Time constant for tank 1 is defined as $\tau_1 = A_1 * R_1$

Time constant for tank 2 is defined as $\tau_2 = A_2 * R_2$

The overall time constant $\tau = \sqrt{\tau_1 \tau_2}$

The damping factor $\xi = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1 \tau_2}}$

$\xi > 1 \Rightarrow \text{Overdamped System}$

$\xi = 1 \Rightarrow \text{Critically damped System (only possible when } \tau_1 = \tau_2)$

System Response:

When we apply as step change (A):

The liquid in tank 1 follows: $h_1(t) = h_{1,ini} - AR_1(1 - e^{-t/\tau_1})$

The liquid in tank 2 follows : $h_2(t) = h_{2,ini} - A\left(1 - \left(1 + \frac{t}{\tau}\right)e^{-t/\tau}\right)$

Apparatus:

1. A reservoir containing the process fluid
2. A constant overhead tank with two outlets (to maintain a steady flow of fluid)
3. Two test tanks each with a single outlet

4. A collection tank (for collecting liquid from the second tank and transferring it again to the reservoir)
5. Linear resistances in the form of a capillary for each outlet.
6. A pump and a regulator system to control to the flow rate
7. Stopwatch
8. Measuring cylinder

Procedure:

1. Ensure all outlet resistances are removed to allow smooth filling of the tanks.
2. Fill the reservoir with a sufficient amount of fluid.
3. Use a pump and regulator system to fill the constant overhead tank while maintaining a steady flow rate.
4. Allow the liquid levels in both tanks to reach steady-state.
5. Verify steady state by ensuring the liquid levels do not fluctuate by more than 0.1-0.2 cm over 2-3 minutes.
6. Record the steady-state liquid levels in both tanks.
7. Measure the input flow rate to Tank 1 after steady state is achieved.
8. At time $t = 0$, decrease the input flow rate to Tank 1 using the regulator.
9. Start the stopwatch simultaneously to track the system's dynamic response.
10. Record the liquid levels in both tanks at regular intervals (e.g., every 30 seconds).
11. Allow the system to reach a new steady state after the step change.
12. Record the final liquid levels and input flow rate for Tank 1.

This process allows us to observe the dynamic response of non-interacting (liquid levels in them) tanks when there is a step input (sudden change in input).

Precautions:

- Ensure all the connections are leak-proof. (Leakage was observed in the experiment).
- Maintain a steady inflow rate to avoid disturbances in the readings.
- Avoid variations in the inlet conditions suddenly.
- Take the readings of the experiment carefully to avoid errors in constant time calculations
- Use precise time and measuring devices to reduce errors and get accurate data.

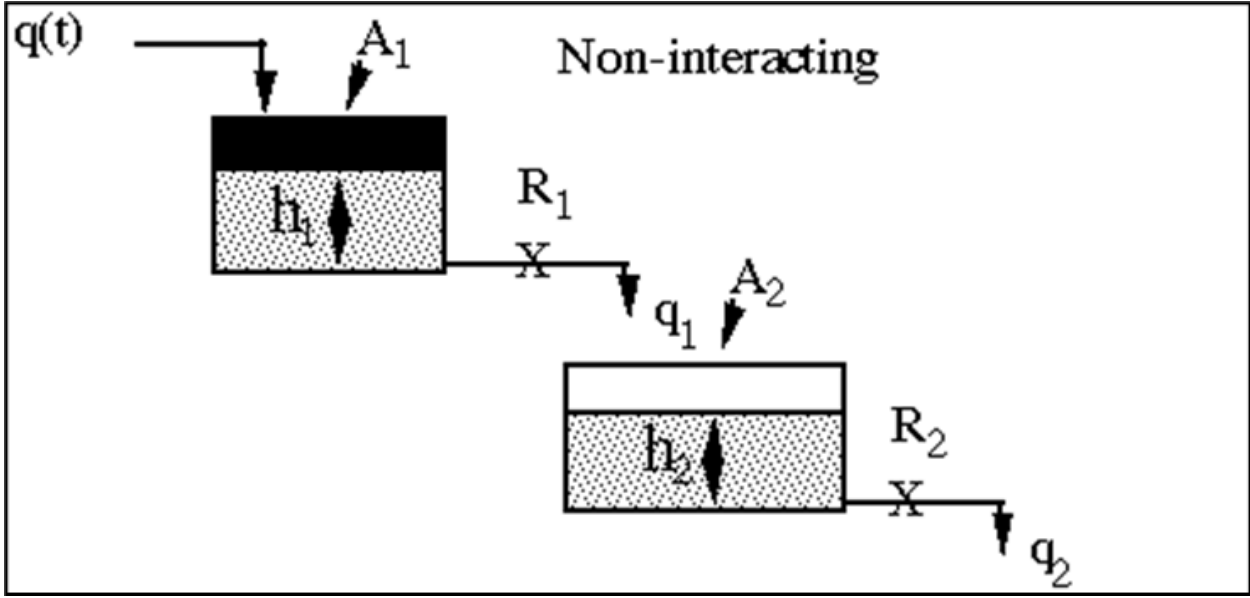


Figure 0: Figure showing the schematic representation of non-interacting tanks set-up

Known parameters:

- Initial steady state height of tank 1 = $h_1 = 17.3 \text{ cm}$
- Initial steady state height of tank 2 = $h_2 = 11.8 \text{ cm}$
- Input flow rate to tank 1 at steady state = $q_1 = 6.6 \frac{\text{ml}}{\text{s}} = 6.6 \frac{\text{cm}^3}{\text{s}}$
- Input flow rate to tank 2 at steady state = $q_2 = 2.5 \frac{\text{ml}}{\text{s}} = 2.5 \frac{\text{cm}^3}{\text{s}}$
- Step input $A = q_s - q_{s1} = 6.6 - 2.5 = 4.1 \frac{\text{cm}^3}{\text{s}}$
- At steady state, the input flow rate of Tank1 = Tank2 $\Rightarrow q(t) = q_1 - q_2 = q_s \frac{\text{cm}^3}{\text{s}}$
- $A_1 = 100 \text{ cm}^2, A_2 = 100 \text{ cm}^2$

Formulae and Calculation:

- Resistance to the flow through tank 1 = $R_1 = \frac{h_1}{q_1} = 2.62 \frac{\text{s}}{\text{cm}^2}$
- Resistance to the flow through tank 2 = $R_2 = \frac{h_2}{q_2} = 1.79 \frac{\text{s}}{\text{cm}^2}$
- The time constant of tank 1 = $\tau_1 = A_1 * R_1 = 262.12 \text{ s}$
- The time constant of tank 2 = $\tau_2 = A_2 * R_2 = 178.79 \text{ s}$
- Theoretical height of tank 1:
 - $h_1(t) = h_{1,ini} - AR_1(1 - e^{-t/\tau_1})$

- Theoretical height of tank 2:

$$\circ h_2(t) = h_{2,ini} - A \left(1 - \left(1 + \frac{t}{\tau} \right) e^{-t/\tau} \right)$$

Where A is the step change given to the system by blocking either one or both inlets of tank 1, t is the experimental time and the overall time constant given by τ .

- $\tau = \sqrt{\tau_1 \tau_2} = 216.48 \text{ s}$
- $\xi = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1 \tau_2}} = 1.02$, ξ is the damping factor

Readings:

TABLE 1: TABLE CONTAINING THE EXPERIMENTAL DATA

S. no.	Time (s)	Observed height of 1st Tank (cm)	Observed height of 2nd Tank (cm)
1.00	0.00	17.30	11.80
2.00	30.00	16.00	11.80
3.00	60.00	15.10	11.80
4.00	90.00	14.00	11.50
5.00	120.00	13.10	11.30
6.00	150.00	12.00	11.10
7.00	180.00	11.20	10.80
8.00	210.00	10.40	10.50
9.00	240.00	9.60	10.10
10.00	270.00	9.00	9.60
11.00	300.00	8.40	9.10
12.00	330.00	7.60	8.60
13.00	360.00	7.30	8.10
14.00	390.00	6.90	7.60
15.00	420.00	6.60	7.00
16.00	450.00	6.00	6.40
17.00	480.00	5.50	5.70
18.00	510.00	5.30	5.00
19.00	540.00	5.00	4.50

20.00	570.00	4.80	3.90
21.00	600.00	4.50	3.10

TABLE 2: TABLE CONTAINING THE CALCULATED DATA

Time (s)	Observed height of 1st Tank (cm)	Theoretical height of 1st Tank (cm)	Observed height of 2nd Tank (cm)	Theoretical height of 2nd Tank (cm)
0.00	17.30	17.30	11.80	11.80
30.00	16.00	16.14	11.80	11.74
60.00	15.10	15.10	11.80	11.57
90.00	14.00	14.18	11.50	11.32
120.00	13.10	13.35	11.30	11.01
150.00	12.00	12.62	11.10	10.68
180.00	11.20	11.96	10.80	10.32
210.00	10.40	11.38	10.50	9.94
240.00	9.60	10.85	10.10	9.57
270.00	9.00	10.39	9.60	9.20
300.00	8.40	9.97	9.10	8.84
330.00	7.60	9.60	8.60	8.50
360.00	7.30	9.27	8.10	8.17
390.00	6.90	8.98	7.60	7.86
420.00	6.60	8.72	7.00	7.57
450.00	6.00	8.48	6.40	7.29
480.00	5.50	8.27	5.70	7.04
510.00	5.30	8.09	5.00	6.80
540.00	5.00	7.92	4.50	6.58
570.00	4.80	7.77	3.90	6.38
600.00	4.50	7.64	3.10	6.20

Results and discussion:

Height of 1st tank (cm) vs Time (s)

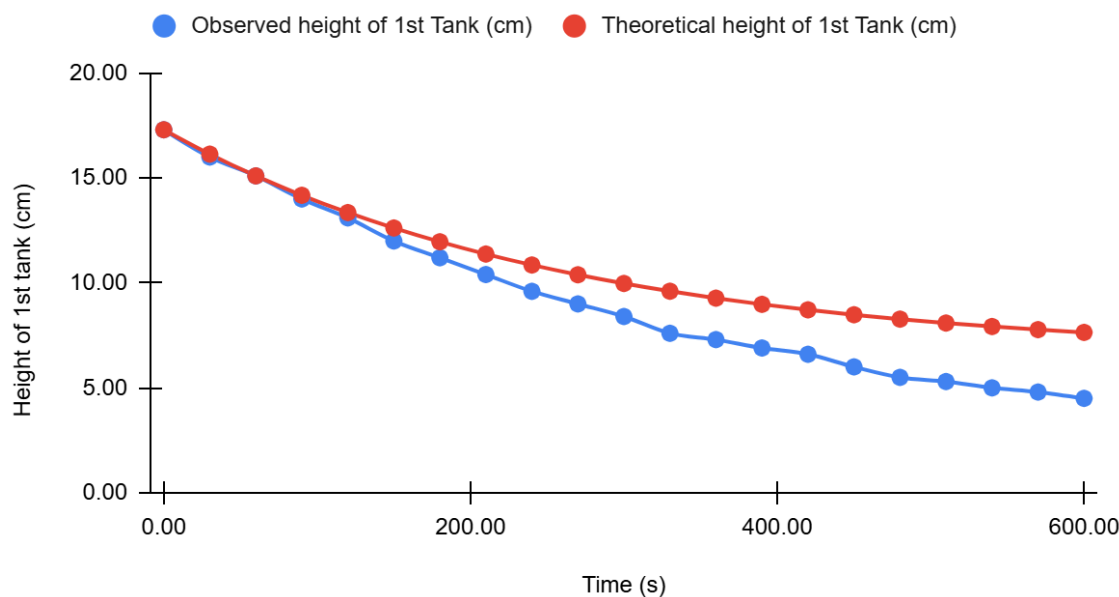


Figure 1: Figure showing the Observed and Theoretical Height of tank 1 (cm) vs time (s)

Height of 2nd tank (cm) vs Time (s)

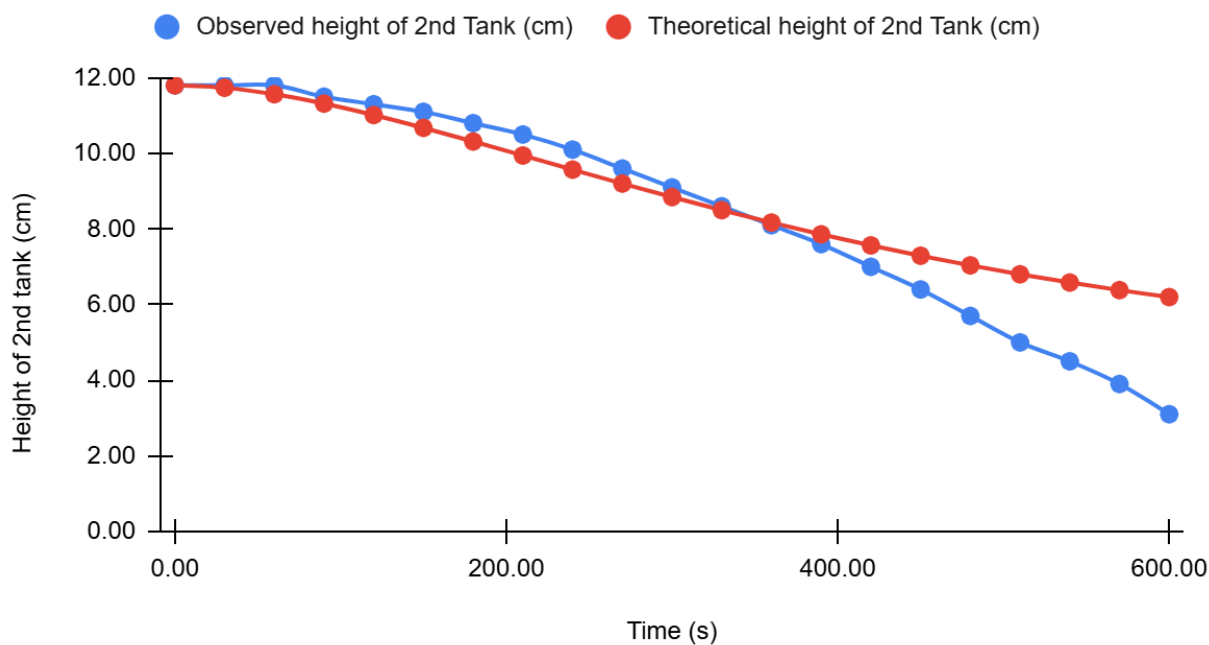


Figure 2: Figure showing the Observed and Theoretical Height of tank 2 (cm) vs time (s)

Observed Results:

$$\tau_1 = 262.12 \text{ s}$$

$$\tau_2 = 178.79 \text{ s}$$

$$\tau = 216.48 \text{ s}$$

$$\text{Damping factor } \xi = 1.02$$

In this experiment the dynamic response of a second-order system comprising two non-interacting tanks was investigated under a step change in inlet flow rate. The experimental observations were compared against theoretical predictions to evaluate the system's behavior, with a focus on time-dependent liquid height in both tanks.

Figure 1 illustrates the variation in the liquid height of Tank 1 over time. The observed height shows a sharp initial decline, followed by a more gradual decrease, ultimately approaching a steady state. This behavior aligns with the expected response of a critically damped first-order system subjected to a step input. The experimental curve accurately captures this trend, with only minor deviations from the theoretical data, especially during the initial transient phase.

The calculated time constant for Tank 1 was $\tau_1 = 262.12 \text{ s}$ consistent with the rapid early response seen in the experiment. The close agreement between theory and observation supports the assumption of linear flow resistance for Tank 1 and confirms that the tank's behavior approximates that of a first-order system. Discrepancies between the observed and predicted values are minimal and can be attributed to minor measurement errors and unavoidable experimental disturbances.

In contrast, the response of Tank 2, shown in Figure 2, exhibited more noticeable deviations from the theoretical prediction. While the theoretical curve suggests a smooth, delayed response due to cascading inflow from Tank 1, the experimental data show a faster rate of decline in liquid level during the later stages of the experiment.

The time constant for Tank 2 was calculated to be $\tau_2 = 178.79 \text{ s}$, leading to an overall system time constant of approximately $\tau = 216.48 \text{ s}$. The damping factor $\xi = 1.02$ further confirms that the system operated in a nearly critically damped regime, which is evident from the absence of oscillations in both plots.

The mismatch between the theoretical and observed curves for Tank 2 can be attributed to several practical factors:

1. **Nonlinear Flow Resistance:** Unlike the theoretical model's assumption of linear resistance, real-world effects such as turbulence and non-uniform flow could introduce nonlinearities, particularly affecting Tank 2's dynamics.

2. **Measurement Uncertainties:** Manual data collection, fluid level reading, and timing could introduce inconsistencies, particularly noticeable in the more gradual response of Tank 2.
3. **Unequal Time Constants:** The theoretical model assumes $\tau_1 = \tau_2$, a condition for critical damping. The experimentally determined unequal time constants invalidate this assumption, leading to an imperfect damping behavior in practice.
4. **Coupled System Dynamics:** As Tank 2 depends on the outflow from Tank 1, any deviation in Tank 1's performance propagates downstream, amplifying discrepancies in Tank 2.

Overall, the absence of oscillatory behavior and the smooth, monotonic decay of liquid height confirm that the system behaves in a near-critically damped manner. The damping factor $\xi = 1.02$ supports this interpretation. The initial steep decline in Tank 1 and the delayed response in Tank 2 are characteristic of a two-stage, cascaded first-order system. Despite minor deviations, the theoretical model provides a reasonable approximation of the system's behavior, particularly for Tank 1.

Conclusion:

This experiment successfully demonstrated the dynamic response of a second-order, critically damped system using a two-tank non-interacting setup subjected to a step input. The results show that Tank 1's behavior closely aligns with theoretical predictions, validating the assumption of first-order system dynamics under ideal flow conditions. In contrast, Tank 2 exhibited a delayed response with noticeable deviations from the theoretical curve, likely due to nonlinear flow resistance, unequal time constants, and experimental inaccuracies.

The system's damping factor, calculated as $\xi = 1.02$, confirms a near-critically damped regime, characterized by a smooth, non-oscillatory transition toward equilibrium. The cascading response of the tanks, with the first reacting immediately to the input and the second lagging due to dependency on outflow, reflects the expected behavior of two first-order systems in series.

While the theoretical model provides a robust approximation, discrepancies in Tank 2 emphasize the need to account for real-world nonlinearities and refine assumptions regarding flow dynamics. The importance of precise measurement and system calibration becomes evident in ensuring the model's predictive accuracy.

Sources of errors:

- There can be small leaks or disturbances in the set up
- There can be minor fluctuations in the inflow rate.

- Since we are using stopwatch and cylinder to measure flow rate there are chances of inaccuracy due to coordination between person keeping stopwatch and person using cylinder.
- We are assuming laminar flow, which may not hold entirely in practical conditions.
- There can be errors when we are taking readings at exactly 30 seconds.

Industrial Relevance:

- Chemical Process Industries for mixing and reaction time control.
- Water Treatment Plants for regulating fluid levels across multiple units.
- Hydraulic Systems where controlled damping and smooth flow transitions are crucial.
- Biochemical Engineering for enzyme kinetics under controlled volumetric conditions.

Academic Relevance:

- Enhances understanding of transient behavior in multi-stage dynamic systems.
- Serves as a practical demonstration of damping characteristics and second-order system responses.
- Reinforces key principles in process control, such as system stability, time constants, and model accuracy.
- Offers foundational insights into the modeling and analysis of liquid-level systems in control engineering curricula.

References:

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- [2] "2nd-order System Dynamics," *controlsystemsacademy.com*. [link](#)
- [3] 'Lab Manual' provided in the classroom.